

Project – PHISHGUARD: INTELLIGENT PROTECTION FROM PHISHING ATTACKS

Introduction

In today's digital landscape, phishing attacks have become one of the most prevalent threats to individuals and organizations alike. These malicious attempts deceive users into revealing sensitive information, such as passwords and financial details, often leading to severe financial loss and data breaches. Traditional security measures, while essential, are not always sufficient to combat the ever-evolving tactics employed by cybercriminals.

The Project

The aim of **PhishGuard** is to develop an AI-driven system to detect phishing emails and websites in real time. The platform will use machine learning algorithms to identify and block phishing attempts before they reach the user.

Goal

To build a system that automatically detects and prevents phishing attacks, minimizing the risks of email fraud and malicious websites.

Prerequisites and Requirements

- Strong understanding of machine learning, email protocols (SMTP, IMAP), and cybersecurity.
- Experience with natural language processing (NLP).

Technologies

- Backend: Python (Django)
- Machine Learning: TensorFlow, Scikit-learn
- Database: MySQL, MongoDB
- Ops: Docker, Kubernetes

Project Milestones

1. Data collection and preprocessing
 - This initial phase involves gathering a diverse dataset of phishing and legitimate emails and websites. The data is cleaned and organized, with important features extracted to ensure it's ready for analysis.
2. Machine learning model for phishing detection
 - A machine learning model is developed using the preprocessed data. Various algorithms are tested to find the most effective one for identifying phishing attempts. The model is trained and validated to optimize its accuracy and reduce false positives.
3. Real-time email and website scanning
 - This milestone focuses on implementing scanning capabilities that analyze incoming emails and accessed websites in real time. The system checks for

suspicious elements to alert users before they engage with potentially harmful content.

4. Threat alerts and reporting

- The system is designed to provide immediate alerts to users when phishing threats are detected. It also includes a reporting feature that logs incidents and performance data, helping refine detection algorithms and offering insights into phishing trends.

5. System integration with existing email platforms

- The final phase involves integrating the PhishGuard system with popular email platforms and web browsers. This ensures seamless functionality and enhances user experience while maintaining robust security measures.